the blood (isotonic with blood). If the Intravenous solutions are too dilute that is hypotonic, the solvent from outside of the cells will flow into the cell to normalise the osmotic pressure and this process which is called hemolysis, causes the cells to burst. On the other hand, if the solution is too concentrated, that is hypertonic, the solvent molecules will flow out of the cells, which causes the cells to shrink and die. For this reason, the Intravenous fluids are prepared such they are isotonic to blood (0.9 % mass/ volume sodium chloride solution).

## 9.10 Reverse osmosis (RO):

Let us consider the experimental setup (Figure 9.15) discussed in the osmosis. The pure water moves through the semipermeable membrane to the NaCl solution due to osmosis. This process can be reversed by applying pressure greater than the osmotic pressure to the solution side. Now the pure water moves from the solution side to the solvent side and this process is called reverse osmosis. It can be defined as a process in which a solvent passes through a semipermeable membrane in the opposite direction of osmosis, when subjected to a hydrostatic pressure greater than the osmotic pressure.

**Application of Reverse osmosis in water purification**:

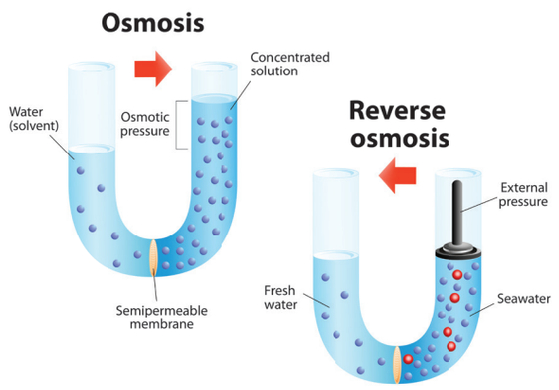



**Figure 9.14 Osmosis & Reverse osmosis**

Reverse osmosis is used in the desalination of sea water and also in the purification of drinking water. A simple set up used in both the process is shown in the figure 9.15. When a pressure higher than the osmotic pressure is applied on the solution side (sea water) the water molecules moves from solution side to the solvent side through semipermeable membrane (Opposite to the Osmotic flow). Pure water can be collected. There are different types of semipermeable membranes used in this process. The membrane used for reverse osmosis has to withstand high pressures. Generally, cellulose acetate or polyamide membranes are commonly used in commercial systems. The selection of membrane used for reverse osmosis will be decided based on the nature of the input water.

### Example Problem-6 :

At 400K 1.5 g of an unknown substance is dissolved in a solvent and the solution is made to 1.5 L. Its osmotic pressure is found to be 0.3 bar. Calculate the molar mass of the unknown substance.

$$\text{Molar mass} = \frac{\text{mass of unknown solute} \times RT}{\text{osmotic pressure} \times \text{volume of solution}}$$

$$= \frac{1.5 \times 8.314 \times 10^{-2} \times 400}{0.3 \times 1.5}$$

$$= 110.85 \text{ gram mol}^{-1}$$

### Evaluate Yourself

13) What is the mass of glucose ($C_6H_{12}O_6$) in one litre solution which is isotonic with 6 g $L^{-1}$ of urea ($NH_2CONH_2$) ?



Unit 9 - jagan.indd 57 27-01-2021 15:06:31